# WILBERT ANDREW YONATHAN

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wilbert-andrew-yonathan.vercel.app | huggingface.co/Wizzas | Sepang, Selangor
*Available for an internship from 29 March 2027 for up to five months*

## EDUCATION

**Xiamen University Malaysia** Sep 2023 – Sep 2027 (Expected)
Bachelor of Engineering in Artificial Intelligence | CGPA: 3.77/4.00

## SKILLS

**Programming Languages:** Python, C, C++, JavaScript, SQL
**Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, LightGBM, Ultralytics
**Computer Vision:** OpenCV, CNNs, YOLOv8, DINOv2, Object Detection, Image Classification, Visual Anomaly Detection
**Generative AI & NLP:** LLMs, RAG, LangChain, LangGraph, Hugging Face Transformers, Qwen, BGE Embeddings
**Data & Databases:** NumPy, pandas, SQLite, Supabase (PostgreSQL)
**Web & APIs:** FastAPI, Flask, React (React.js), Gradio, Streamlit, REST APIs
**Cloud & Deployment:** AWS (Lightsail, EC2, Lambda, CloudWatch, Budgets), Docker, Docker Compose, Linux (Ubuntu), Git, Hugging Face Spaces, Render, Vercel
**Automation & Integration:** n8n, Webhooks, Resend, Formspree
**Spoken Languages:** English, Mandarin, Indonesian, Malay, Hokkien

## PROJECT EXPERIENCE

**PaddyLiteX: Paddy Disease Detection (Final-Year Thesis)** Apr 2026 – Present
- Developed a DenseLiteX-SPPF-C2PSA YOLOv8n detector for nine paddy classes, achieving 81.67% mAP@0.5 on 803 test images (+0.88 percentage points and 4.24% fewer parameters than the YOLOv8n baseline).
- Reduced parameters by 68.01% and GFLOPs by 72.59% relative to the initial YOLOv8n-DenseNet121 proposal; conducted ablation studies and deployed a three-model comparison demo.

**DefectRAG: Visual Anomaly Inspection and RAG** Aug 2026 – Oct 2026
- Built and evaluated a visual anomaly-inspection pipeline with a frozen DINOv2 encoder across 12 VisA product categories, achieving 95.13% mean image AUROC in a retrospective development evaluation.
- Integrated BGE embeddings, Qwen, LangChain and LangGraph to generate cited RAG explanations; developed a React.js/FastAPI dashboard, packaged locally with Docker, and deployed a Hugging Face ZeroGPU demo.
- Built a corrective-action workflow with Supabase (Auth, row-level security, TOTP MFA) and n8n: instant supervisor emails, reminders, duplicate-send protection; deployed on AWS Lightsail with Docker Compose and Caddy (HTTPS).

**MarineLite: Marine Organism Detection** Jun 2026 – Jul 2026
- Integrated GhostConv and SimSPPF into DU-MobileYOLO, improving mAP@0.5 from 74.23% to 75.36% on URPC2020 while reducing parameters from 4.70M to 4.37M.
- Conducted ablation studies and deployed a web demo with detection overlays and model comparisons.

**dPickleball: Reinforcement Learning Agent** May 2026 – Jul 2026
- Trained Proximal Policy Optimization (PPO) agents in Unity ML-Agents using curriculum learning, reward shaping and league self-play.
- Contributed candidate policies and round-robin evaluation to a six-member team that placed 2nd among 13 teams in the final class competition.

## WORK EXPERIENCE

**Student Registration & Operations Assistant, Xiamen University Malaysia** Apr 2025
- Supported registration logistics and directed students within a team serving 800+ students.
- Organised university assets and materials for event operations and storage.

## LEADERSHIP AND ACTIVITIES

**Head of Photography & Videography, Love in Action** Feb 2026 – Mar 2026
Coordinated media coverage and edited outreach materials.

**Head of General Affairs, XMUM AI Club** Apr 2025 – Oct 2025
Coordinated logistics and external partners for a national-level AI event.

**Assistant, General Affairs, AIESEC XMUM** Sep 2023 – Sep 2025
Coordinated room bookings and venue setups for weekly meetings.

## AWARDS

- Dean's List (three consecutive semesters)
- XMUM Merit Scholarship (four-year undergraduate programme, 2023–2027)

## REFERENCES

**Shaidah Binti Jusoh**
Assistant Professor | Thesis Supervisor
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**Goh Sim Kuan**
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